

ACME ERP • QA TEST SUITE

End-to-End Manual Test Cases

Unit Types & Items to PO to Inventory to POS Dispensary Lifecycle

Target System: ACME ERP
v4.2

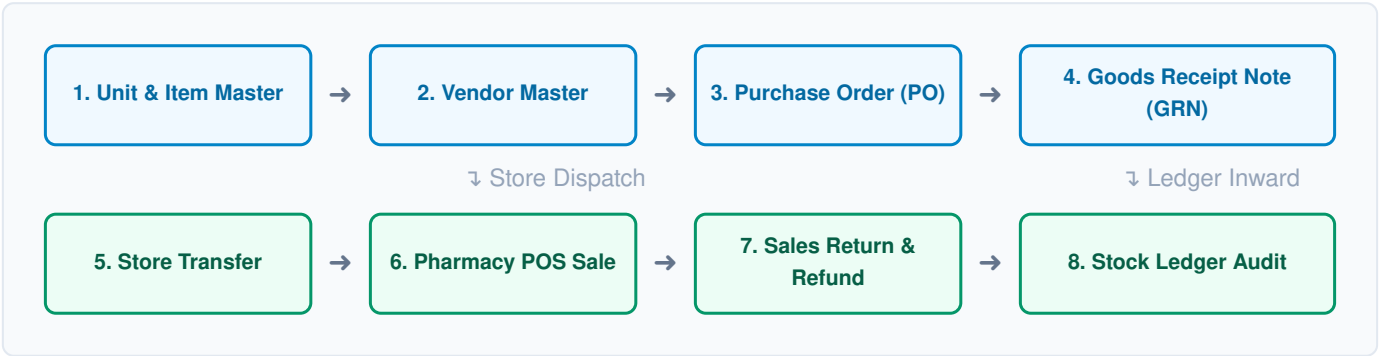
Category: Procurement &
Inventory

Scope: Master Data to POS &
Ledger

Execution: Manual
Acceptance Test

This test suite covers the complete procurement, inventory lifecycle, and point-of-sale dispensing workflow in **ACME ERP**. It validates multi-tier unit conversions, vendor PO generation, automated taxation, GRN inwarding, inter-store requisition transfers, FEFO batch-wise dispensing at pharmacy POS, customer sales returns, and real-time stock ledger auditing.

Workflow Overview Diagram



Test Suite Summary

MODULE / PHASE	TEST CASE ID	TEST SCENARIO	PRIORITY
1. Unit & Item Masters	TC-MST-01	Create Base Units, Secondary Units & Conversion Multipliers	HIGH
	TC-MST-02	Create Item with Unit Pricing Matrix, GST %, and Reorder Levels	HIGH
	TC-MST-03	Verify Unit Conversion Multiplier Calculations on Item Profile	MEDIUM
2. Vendor Management	TC-VND-01	Create Vendor with GSTIN, Contact & Payment Terms	MEDIUM
3. Purchase Orders (PO)	TC-PO-01	Create Multi-Item PO with Auto GST & Line-Item Calculations	HIGH
	TC-PO-02	Verify PO Number Auto-Generation (PO-YYYY-XXXX)	MEDIUM
	TC-PO-03	Record Partial / Full Advance Payment against PO	HIGH
4. Goods Receipt Note (GRN)	TC-GRN-01	Inward Goods against PO with Batch, Expiry & Free Qty	CRITICAL
	TC-GRN-02	Direct GRN Inwarding (Without PO)	HIGH
	TC-GRN-03		HIGH

MODULE / PHASE	TEST CASE ID	TEST SCENARIO	PRIORITY
		Partial GRN Receiving & PO Status Auto-Transition (<code>partial</code> → <code>closed</code>)	
	<code>TC-GRN-04</code>	Stock Ledger & Store Batch Stock Inward Verification	CRITICAL
5. Purchase Invoices & Bills	<code>TC-PINV-01</code>	Convert GRN to Purchase Invoice & Settle Vendor Payable	HIGH
6. Store Transfers & Indents	<code>TC-TRN-01</code>	Create Inter-Store Requisition from Pharmacy to Central Store	HIGH
	<code>TC-TRN-02</code>	Fulfill Requisition & Issue Stock Transfer (<code>in_transit</code>)	HIGH
	<code>TC-TRN-03</code>	Receive Stock Transfer at Destination Store & Verify Balances	CRITICAL
7. Stock Adjustments	<code>TC-ADJ-01</code>	Physical Audit Reconciliation (Gain / Loss / Expiry Write-Off)	MEDIUM
8. Pharmacy POS Billing	<code>TC-POS-01</code>	Walk-in Patient Sale with Barcode Scan & FEFO Auto-Select	CRITICAL
	<code>TC-POS-02</code>	Prescription-Linked Billing with Doctor & Patient Selection	HIGH
	<code>TC-POS-03</code>	Split / Mixed Payment Processing (Cash + UPI)	HIGH
	<code>TC-POS-04</code>	Out-of-Stock / Insufficient Quantity Validation	HIGH
9. Sales Returns & Refunds	<code>TC-RET-01</code>	Process Partial Sales Return with Cash Refund & Stock Restoration	HIGH
10. Ledger & Audit Trail	<code>TC-AUD-01</code>	End-to-End Stock Ledger Verification (<code>stockLedger</code> FIFO/FEFO trace)	CRITICAL

Detailed Test Cases

Phase 1: Unit Types, Conversions & Item Master Setup

TC-MST-01 | Create Base Units, Secondary Units & Conversion Multipliers

HIGH PRIORITY

Preconditions: Logged in with `admin` role.

Navigation: Sidebar → `Purchases` → `Unit Types` (`/purchases/unit-types`).

Test Steps:

1. Click "New Unit Type".
2. Create Base Unit: Name = `Tablet`, Symbol = `TAB`, Category = `Count/Quantity`, Check "Base Unit" = `Yes`. Click **Save**.
3. Create Secondary Unit: Name = `Strip of 10`, Symbol = `STRIP-10`, Category = `Count/Quantity`, Check "Base Unit" = `No`. Click **Save**.
4. Create Packaging Unit: Name = `Box of 10 Strips`, Symbol = `BOX-100`, Category = `Count/Quantity`, Check "Base Unit" = `No`. Click **Save**.
5. Go to "Unit Conversions" tab.
6. Add Conversion 1: `1 STRIP-10` = `10 TAB` (Multiplier = `10`).
7. Add Conversion 2: `1 BOX-100` = `10 STRIP-10` (Multiplier = `10`).

✓ EXPECTED RESULTS

- All unit types are listed with appropriate base unit badges.
- Conversion matrix validates forward and reverse multiplication factors without circular errors.

TC-MST-02 | Create Item with Multi-Unit Pricing, GST %, and Reorder Levels

HIGH PRIORITY

Preconditions: Units from `TC-MST-01` exist. Item Type `Medicines` exists.

Navigation: Sidebar → `Purchases` → `Items` (`/purchases/items`).

Test Steps:

1. Click "Add New Item".
2. Enter Details:
 - **Item Name:** `Paracetamol 500mg (Acme)`
 - **Category / Item Type:** `Medicines`
 - **Base Unit:** `Tablet (TAB)`
 - **Purchase Unit:** `Box of 10 Strips (BOX-100)`
 - **Sale Unit:** `Strip of 10 (STRIP-10)`
 - **HSN Code:** `30049099`
 - **GST Rate:** `12%`
 - **Purchase Rate (per Purchase Unit):** `₹ 250.00`
 - **Sale Price (per Sale Unit):** `₹ 35.00`
 - **Reorder Level:** `50`
 - **Reorder Quantity:** `100`
 - **Drug Schedule:** `Schedule H`
3. Click "Save Item".

✓ EXPECTED RESULTS

- Item is created with unique ID and appears in items directory.
- Unit price matrix calculates base cost per tablet (`₹ 2.50`) and retail price per tablet (`₹ 3.50`).

Phase 2: Vendor Management

TC-VND-01 | Create Supplier / Vendor

MEDIUM PRIORITY

Navigation: Sidebar → Purchases → Vendors (/purchases/vendors).

Test Steps:

1. Click "Add Vendor".
2. Enter Vendor Details:
 - **Vendor Name:** MedPharma Distributors Ltd
 - **GSTIN:** 27AABCU9603R1ZM (Valid 15-character GSTIN)
 - **Phone:** 9876543210
 - **Contact Person:** Rajesh Kumar
 - **Address:** Plot 42, MIDC Pharma Zone, Mumbai
3. Click "Save".

✓ **EXPECTED RESULTS**

- Vendor saved successfully with active status.
- Vendor is immediately searchable in PO vendor dropdown.

Phase 3: Purchase Order (PO) Creation & Payment

TC-PO-01 | Create Multi-Item Purchase Order with Automated Tax Calculation

HIGH PRIORITY

Preconditions: Vendor MedPharma Distributors Ltd and Item Paracetamol 500mg exist.

Navigation: Sidebar → Purchases → Purchase Orders → New PO (/purchases/purchase-orders/new).

Test Steps:

1. Select Vendor: MedPharma Distributors Ltd .
2. Set **PO Date:** Current Date.
3. Add Line Item 1:
 - Item: Paracetamol 500mg (Acme)
 - Unit: Box of 10 Strips (BOX-100)
 - Ordered Qty: 20
 - Unit Rate: ₹ 250.00
 - GST %: 12%
 - Expected Line Value: $20 * 250 = ₹ 5,000.00 + 12\% \text{ GST } (₹ 600.00) = ₹ 5,600.00$.
4. Enter Remarks: Urgent stock replenishment for central pharmacy .
5. Click "Create Purchase Order".

✓ **EXPECTED RESULTS**

- PO created with auto-generated code PO-2026-XXXX .
- PO Status is marked "open" and Payment Status is marked "unpaid".
- Total PO Amount displays ₹ 5,600.00 .

TC-PO-03 | Record Advance Payment against Purchase Order

HIGH PRIORITY

Preconditions: TC-PO-01 completed.

Navigation: View PO Details (`/purchases/purchase-orders/$id`).

Test Steps:

1. Click **"Record Payment"** button.
2. Amount: ₹ 2,000.00 (Partial advance).
3. Payment Mode: Bank Transfer (RTGS/NEFT) .
4. Reference No: UTR9928374112 .
5. Payment Date: Today.
6. Click **"Save Payment"**.

✓ EXPECTED RESULTS

- Payment entry is logged in PO payment history table.
- PO Payment Status updates from **"unpaid"** to **"partial"**.
- Remaining balance displays ₹ 3,600.00 .

Phase 4: Goods Receipt Note (GRN) & Stock Inwarding

TC-GRN-01 | Inward Goods against PO with Batch, Expiry & Unit Conversion

CRITICAL

Preconditions: TC-PO-01 created. Store Central Medical Store exists.

Navigation: Sidebar → Purchases → GRN → New GRN (`/purchases/grns/new`).

Test Steps:

1. Select **PO Number:** Select PO-2026-XXXX .
2. Select **Destination Store:** Central Medical Store .
3. Verify PO items auto-populate into GRN line table.
4. Fill Inward Item Details:
 - Received Qty: 20 (Boxes)
 - Free Qty: 2 (Boxes promo)
 - Batch Number: PARA-2026-B1
 - Expiry Date: 2028-06-30 (Future date)
 - Unit Rate: ₹ 250.00
 - Sale Price (per Strip): ₹ 35.00
5. Enter Challan / Delivery Note No: DC-98441 .
6. Set Status: **"Posted"** (or Submit & Post).
7. Click **"Save & Post GRN"**.

✓ EXPECTED RESULTS

- GRN created with sequential code GRN-2026-XXXX .
- PO status transitions from **"open"** to **"closed"** (all ordered qty received).
- Stock is atomically incremented in `inventory.store_batch_stock` for Central Medical Store .
- Total received units converted to base units (e.g., 22 boxes * 10 strips = 220 strips or 2,200 tablets).
- Audit trail entry written to `inventory.stock_ledger` with movement type GRN .

TC-GRN-03 | Partial GRN Receiving Workflow

HIGH PRIORITY

Preconditions: Create a new PO for 50 boxes.

Test Steps:

1. Create GRN against this PO receiving only 20 boxes out of 50.
2. Post the GRN.

✓ EXPECTED RESULTS

- PO Status updates to "partial".
- Subsequent GRN form displays remaining pending quantity = 30 boxes.

Phase 5: Inter-Store Stock Requisitions & Transfers

TC-TRN-01 | Requisition Indent from Retail Pharmacy to Central Store

HIGH PRIORITY

Preconditions: Stock exists in Central Medical Store.

Navigation: Sidebar → Inventory → Requisitions → New Requisition (/inventory/requisitions/new).

Test Steps:

1. From Store (Requesting): Retail Pharmacy.
2. To Store (Supplying): Central Medical Store.
3. Priority: Normal.
4. Item: Paracetamol 500mg (Acme).
5. Requested Quantity: 50 (Strips).
6. Purpose: Daily dispensing buffer refill.
7. Click "Submit Requisition".

✓ EXPECTED RESULTS

- Requisition generated with code SREQ-2026-XXXX and status "submitted".

TC-TRN-02 & TC-TRN-03 | Fulfill Requisition, Stock Transfer Issue & Receipt

CRITICAL

Navigation: Sidebar → Inventory → Transfers → New Transfer (/inventory/transfers/new).

Test Steps:

1. Select Source: Central Medical Store, Destination: Retail Pharmacy.
2. Select Requisition: SREQ-2026-XXXX.
3. Select Batch: PARA-2026-B1 (Shows available stock).
4. Transfer Qty: 50 strips.
5. Click "Dispatch / Issue Stock".
6. Verify Transfer Status becomes "in_transit".
7. Switch / Log in as Retail Pharmacy In-charge.
8. Open Transfer STRN-2026-XXXX and click "Confirm Receipt".

✓ EXPECTED RESULTS

- Central Medical Store stock decreases by 50 strips (TRANSFER_OUT in ledger).
- Retail Pharmacy stock increases by 50 strips (TRANSFER_IN in ledger).
- Batch PARA-2026-B1 is now active and sellable in Retail Pharmacy.

Phase 6: Pharmacy Point-of-Sale (POS) Billing

TC-POS-01 | Walk-in Patient Quick Sale with Auto-FEFO Batch Selection

CRITICAL

Preconditions: Stock available in **Retail Pharmacy** (**50** strips of **PARA-2026-B1**).

Navigation: Sidebar → **Inventory** → **Point of Sale (POS)** (**/inventory/pos**).

Test Steps:

1. Select Active Store: **Retail Pharmacy** .
2. Customer / Patient Type: **Walk-in** .
3. Customer Name: **John Doe** , Phone: **9811223344** .
4. In Search bar, type **Paracetamol** or scan barcode.
5. Select **Paracetamol 500mg (Acme)** .
6. Verify the system auto-picks batch **PARA-2026-B1** with earliest expiry.
7. Set Qty: **2** (Strips) @ **₹ 35.00** per strip.
8. Verify Subtotal = **₹ 70.00** .
9. Add 5% Line Discount (**₹ 3.50**).
10. Net Amount = **₹ 66.50** .
11. Select Payment Method: **Cash** . Enter Tendered Amount = **₹ 100.00** .
12. Verify Change Return = **₹ 33.50** .
13. Click **"Complete Sale & Print Receipt"**.

✓ EXPECTED RESULTS

- Sales invoice created with sequential number **SINV-2026-XXXX** .
- Thermal / PDF receipt modal triggers for printing.
- Retail Pharmacy batch stock decreases from **50** to **48** strips.
- **stockLedger** logs a **SALE** entry of **-2** strips.

TC-POS-02 | Prescription-Linked Sale with Doctor & OPD Encounter

HIGH PRIORITY

Preconditions: Patient **Anjali Sharma** registered with doctor prescription.

Navigation: **POS** (**/inventory/pos**).

Test Steps:

1. Select **"Prescription Lookup"**.
2. Search by Patient Name or MRN.
3. Select prescription: Prescribed medicines auto-populate into billing cart with exact dosages.
4. Select Payment Method: **UPI** (Enter Txn ID / QR scan).
5. Click **"Complete Sale"**.

✓ EXPECTED RESULTS

- Sale successfully tied to Patient ID and Doctor ID.
- Prescription marked as dispensed/completed.

TC-POS-03 | Split / Mixed Payment Processing

HIGH PRIORITY

Navigation: POS (/inventory/pos).

Test Steps:

1. Add items totaling ₹ 1,500.00 .
2. Select Payment Mode: "Mixed".
3. Split 1: Cash = ₹ 500.00 .
4. Split 2: Card = ₹ 1,000.00 (Auth Code: TXN88392).
5. Click "Complete Sale".

✓ EXPECTED RESULTS

- Invoice logs both payment lines with respective payment channels.
- Handover and daily cash closing correctly account for ₹500 in physical cash drawer and ₹1,000 in bank POS terminal.

TC-POS-04 | Negative / Out-of-Stock Validation

HIGH PRIORITY

Navigation: POS (/inventory/pos).

Test Steps:

1. Add an item with available stock = 5 strips.
2. Attempt to input quantity = 10 strips.
3. Click "Complete Sale".

✓ EXPECTED RESULTS

- System blocks checkout with clear validation error: "Requested quantity (10) exceeds available stock (5) for Batch XXX".
- No negative stock is allowed in the ledger.

Phase 7: Sales Returns & Credit Notes

TC-RET-01 | Process Partial Sales Return with Stock Restoration

HIGH PRIORITY

Preconditions: Invoice SINV-2026-XXXX from TC-POS-01 completed.

Navigation: Sidebar → Inventory → Invoices / Returns (/inventory/invoices).

Test Steps:

1. Search and open invoice SINV-2026-XXXX .
2. Click "Initiate Return / Credit Note".
3. Select Item: Paracetamol 500mg .
4. Return Qty: 1 (Strip).
5. Return Reason: Unopened medicine returned by patient .
6. Refund Mode: Cash .
7. Click "Process Refund & Restock".

✓ EXPECTED RESULTS

- Return voucher generated with status "completed".
- Stock for PARA-2026-B1 in Retail Pharmacy restores from 48 back to 49 strips.
- Stock ledger creates a POS_RETURN entry of +1 strip.

Phase 8: Stock Ledger Audit & Reconciliations

TC-AUD-01

End-to-End Stock Ledger Audit Trail Verification

CRITICAL

Navigation: Sidebar → `Inventory` → `Stock Ledger` (`/inventory/ledger`).

Test Steps:

- Filter by Item: `Paracetamol 500mg (Acme)`.
- Filter by Date: Today.
- Observe chronological sequence of movements:
 - Entry 1: `GRN` (Inward `+220` strips into Central Store).
 - Entry 2: `TRANSFER_OUT` (`-50` strips from Central Store).
 - Entry 3: `TRANSFER_IN` (`+50` strips into Retail Pharmacy).
 - Entry 4: `SALE` (`-2` strips from Retail Pharmacy).
 - Entry 5: `POS_RETURN` (`+1` strip into Retail Pharmacy).
- Verify running balance columns match actual physical stock:
 - `Central Medical Store` : `170` strips.
 - `Retail Pharmacy` : `49` strips.
 - Total Entity Stock: `219` strips.

✓ **EXPECTED RESULTS**

- Zero discrepancies across all stores and transaction types.
- Every ledger entry is mapped to a valid document number and user ID.

Boundary, Security & Edge Cases Matrix

CASE ID	SCENARIO	INPUT / ACTION	EXPECTED BEHAVIOR
TC-SEC-01	Role-based store isolation	Non-pharmacist staff attempting POS sale	Access restricted (<code>403 Forbidden</code> / store selection hidden).
TC-EDG-01	Expired batch dispensing	Attempting to bill batch with <code>expiryDate < today</code>	POS prevents selection and marks batch as expired.
TC-EDG-02	Fractional unit dispensing	Prescribing <code>0.5</code> strip or liquid measurement	Supported only when <code>allowFractional = true</code> on Item master.
TC-EDG-03	Concurrent checkout race	Two counters selling last batch unit simultaneously	Database transaction locks stock row; second transaction fails gracefully with out-of-stock notification.